
Xue Zheng

Ph.D. Candidate, Electrical and Computer Engineering
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Research Interests

Federated and decentralized learning; distributed optimization; privacy-preserving and robust machine learning; communication-efficient learning; foundation models and mixture-of-experts systems.

Education

The Ohio State University, Columbus, OH
Ph.D. in Electrical and Computer Engineering
Advisor: Prof. Aylin Yener
Candidacy Examination: Completed April 2025

Jan. 2022–Present
Expected Dec. 2026

The Ohio State University, Columbus, OH
M.S. in Electrical and Computer Engineering
Advisor: Prof. Parinaz Naghizadeh

Jan. 2020–Dec. 2021

Harbin Institute of Technology at Weihai, Weihai, China
B.Eng. in Information and Electrical Engineering
Honor: Merit Student of Harbin Institute of Technology

Sept. 2014–June 2018

Research Experience

Graduate Research Assistant, The Ohio State University
Advisor: Prof. Aylin Yener

Jan. 2022–Present

Performative and Distribution-Shift-Aware Learning:

EPIC-DPL: Decentralized Performative Learning under Coupled Graph and Data Evolution

- Formulated decentralized performative prediction over random, directed, and time-varying peer-interaction graphs, where deployed models induce the local data distributions used in subsequent updates.
- Introduced separate deployment and graph-update periods to characterize the coupled effects of intermittent model deployment, peer-interaction evolution, and performative data shifts.
- Developed EPIC-DPL, a fully decentralized stochastic-gradient method with a randomized common-outdegree peer-selection protocol that preserves the network average in expectation.
- Established convergence to the global performative stable point and characterized how deployment frequency, graph-update frequency, and peer-interaction density affect consensus, convergence, and steady-state accuracy.

ProFL: Performative Robust Optimal Federated Learning

- Developed a robust federated learning method for model-induced distribution shifts under noisy and contaminated data.
- Targeted the performative optimal solution rather than performative stable points.
- Established convergence guarantees under the Polyak–Łojasiewicz condition for non-convex objectives.
- Demonstrated improved performance over existing performative federated learning methods across multiple datasets.

Federated Learning Under Evolving Distribution Shifts

- Studied federated learning when client data distributions evolve over time and models must generalize to unseen future domains.
- Contributed to the development and experimental evaluation of FedEvolve and FedEvp, including experiments on synthetic and real datasets under evolving shifts and client heterogeneity.

Federated Large-Model Fine-Tuning and Resource-Constrained Deployment:

S^3 -MoE: Server-Side Sparse Assignment for Federated Mixture-of-Experts

- Studied federated MoE learning under strict partial expert visibility, where resource-constrained clients can access only a limited subset of experts.
- Developed a server-side expert assignment framework that estimates client-specific expert relevance and iteratively refines subset allocation through an exploit–explore mechanism.
- Established bounded relevance estimation error and exact top- K expert recovery guarantees under partial observations and evolving model parameters.
- Demonstrated on CIFAR-100 that the proposed method captures client-specific expert specialization and outperforms full-expert-loading sparse activation baselines while using fewer locally available experts.

Federated Fine-Tuning of Large Language Models with Parameter-Efficient Methods

- Implemented and evaluated federated fine-tuning methods for large language models, including naive LoRA, Hybrid LoRA, and Zeroth-Order method.
- Focused on Hybrid LoRA, where client-specific LoRA parameters remain local while the shared base model is periodically synchronized across clients, enabling partial personalization and reducing exposure of local adaptation parameters.
- Contributed to the implementation, debugging, hyperparameter tuning, and experimental evaluation of Hybrid LoRA and Zeroth-Order method on a multi-GPU server with four NVIDIA H100 GPUs.
- Developed and tested an AWS-based communication and aggregation pipeline for multi-client federated fine-tuning experiments.
- Collaborated across multiple research groups to integrate, evaluate, and present federated LLM fine-tuning workflows in a multi-client setting.

Personalized, Decentralized, and Privacy-Preserving Peer-to-Peer Learning:

DiPLe: Learning Directed Collaboration Graphs for Peer-to-Peer Personalized Learning

M.S. and Ph.D. research

Advisors: Prof. Parinaz Naghizadeh and Prof. Aylin Yener

- Developed a fully decentralized personalized learning method for agents with non-IID local datasets.
- Enabled each agent to identify relevant collaborators and jointly learn a weighted, directed collaboration graph together with its personalized model.
- Established generalization error bounds and convergence guarantees for the proposed algorithm.
- Demonstrated improved personalization performance over existing federated and personalized learning baselines.

Privacy-Preserving Clustered Peer-to-Peer Learning

- Extended peer-to-peer federated learning to privacy-sensitive collaboration without relying on a central server.
- Used client clustering to make Shamir-secret-sharing-based aggregation practical over denser peer-to-peer communication graphs, reducing communication overhead.
- Integrated Shamir secret sharing within each cluster so that clients can securely aggregate local models without revealing any individual client’s model to its peers.
- Established convergence guarantees and demonstrated improved accuracy and communication efficiency compared with existing privacy-preserving federated learning baselines.

Earlier Research:

Undergraduate Researcher, Harbin Institute of Technology at Weihai Jan. 2018–June 2018

- Studied quantitative photoacoustic imaging using sparse decomposition.
- Implemented K-SVD dictionary learning and orthogonal matching pursuit in MATLAB.

Publications and Manuscripts

Published and Accepted

1. **X. Zheng**, T. Xie, X. Tan, A. Yener, and X. Zhang, “ProFL: Performative Robust Optimal Federated Learning,” accepted for publication in *IEEE Transactions on Signal Processing*, 2026.
2. **X. Zheng** and A. Yener, “Learning What to Load: Finite-Profile Stability of Task-Conditioned Resident Experts,” accepted for poster presentation at *DeepMath 2026*.
3. **X. Zheng** and A. Yener, “Server-Side Sparse Assignment for Federated Mixture-of-Experts,” accepted to the *2026 IEEE Information Theory Workshop (ITW)*, 2026.
4. Z. Luo et al., including **X. Zheng**, “Democratizing LLM Training Across Geo-Distributed Heterogeneous Compute: A Systems Position and Roadmap,” long paper accepted for poster presentation at the *NeurIPS Workshop on Collaborative, Open, and Decentralized Training of Foundation Models (CODEC-FM)*, 2026.
5. Z. Luo, J. Cai, C. Le Denmat, S. Nair, F. Nourzad, R. K. Sudha, Q. Wu, J. Zhang, S. Lee, Z. Li, P. Qiu, S. Shah, R. Sharma, S. Srinivasan, Y. Xia, **X. Zheng**, Z. Liu, B. Ying, K. Chowdhury,

G. Joshi, Y. Liang, R. Nowak, S. Parthasarathy, S. Prakash, B. Ravindran, S. Shakkottai, N. B. Shroff, S. Srinivasan, H. Yang, A. Yener, and J. Liu, “Toward WAN-Aware LLM Training Across Heterogeneous, Geo-Distributed Sites,” in *Proceedings of the ACM SIGCOMM Workshop on Networks for AI Computing (NAIC)*, 2026.

6. X. Tan, T. Xie, **X. Zheng**, A. Yener, M. Lee, A. Payani, H. Latapie, and X. Zhang, “Federated Learning Under Evolving Distribution Shifts,” *Entropy*, vol. 28, no. 1, Art. no. 101, 2026. doi:10.3390/e28010101
7. **X. Zheng** and A. Yener, “Privacy-Preserving Clustered Peer-to-Peer Learning,” in *Proceedings of the 2023 57th Asilomar Conference on Signals, Systems, and Computers*, pp. 704–708, 2023.
8. **X. Zheng**, P. Naghizadeh, and A. Yener, “DiPLe: Learning Directed Collaboration Graphs for Peer-to-Peer Personalized Learning,” in *Proceedings of the 2022 IEEE Information Theory Workshop (ITW)*, pp. 446–451, 2022.

Manuscript in Revision

1. **X. Zheng** and A. Yener, “Decentralized Multi-Agent Performative Prediction under Coupled Graph and Deployment-Induced Data Evolution,” manuscript in revision, 2026.

Teaching and Mentoring Experience

Lecturer and Graduate Student Mentor, AI-EDGE Summer REU Program Summers 2024–2026

- **Summer 2024:** Participated in an undergraduate research project on generative modeling. Delivered lectures on fundamental probability concepts and mentored students on research problem formulation, project development, and improving the presentation of their work.
- **Summer 2025:** Delivered a lecture on fundamental probability concepts to support undergraduate students preparing for research projects in artificial intelligence and machine learning.
- **Summer 2026:** Delivered an updated lecture connecting probability concepts with modern generative AI, including probabilistic token generation and the role of temperature in large language model outputs.

Grader, Computer Architecture and Design, The Ohio State University Aug. 2021–Dec. 2021

- Graded homework and machine problems and assisted students during office hours.

Presentations and Posters

- Z. Luo et al., including **X. Zheng**, “Democratizing LLM Training Across Geo-Distributed Heterogeneous Compute: A Systems Position and Roadmap,” poster accepted for presentation at the *NeurIPS Workshop on Collaborative, Open, and Decentralized Training of Foundation Models (CODEC-FM)*, 2026.
- X. Zheng and A. Yener, “Learning What to Load: Finite-Profile Stability of Task-Conditioned Resident Experts,” poster accepted for presentation at DeepMath 2026.
- X. Zheng, T. Xie, X. Tan, A. Yener, and X. Zhang, “ProFL: Performative Robust Federated Learning,” poster presented at the Artificial Intelligence Modeling, Analysis, and Control of Complex Systems (AIMACCS) Workshop, Columbus, OH, June 1–2, 2026.
- X. Zheng and A. Yener, “Privacy-Preserving Clustered Peer-to-Peer Learning,” poster presented at the Asilomar Conference on Signals, Systems, and Computers, Pacific Grove, CA, Oct. 29–Nov. 1, 2023.
- X. Zheng, P. Naghizadeh, and A. Yener, “DiPLe: Learning Directed Collaboration Graphs for Peer-to-Peer Personalized Learning,” oral presentation delivered virtually at the 2022 IEEE Information Theory Workshop (ITW), 2022.

Professional Service

- Reviewer, International Conference on Artificial Intelligence and Statistics (AISTATS 2026), 2025.
- Reviewer, IEEE Transactions on Signal Processing, 2025–2026.
- Reviewer, IEEE Journal of Selected Topics in Signal Processing, 2024–2026.

Technical Skills

Programming: Python, MATLAB, C, Java; coursework experience with CUDA C

Machine Learning and Data Science: PyTorch, Hugging Face Transformers, TensorFlow, NumPy, scikit-learn, pandas, Matplotlib, Jupyter

Large-Model and Distributed Training: Megatron-LM

Development and Computing: Git, GitHub, Linux, Docker, SSH, remote GPU servers

Technical Communication: LaTeX, Overleaf, Microsoft PowerPoint, scientific figure and presentation design